[INCH-POUND]

MIL-C-39012/133
27 June 1989

MILITARY SPECIFICATION SHEET

CONNECTOR, RECEPTACLE, ELECTRICAL, COAXIAL, RADIO FREQUENCY, SERIES SMC, PRINTED CIRCUIT BOARD MOUNT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-39012.

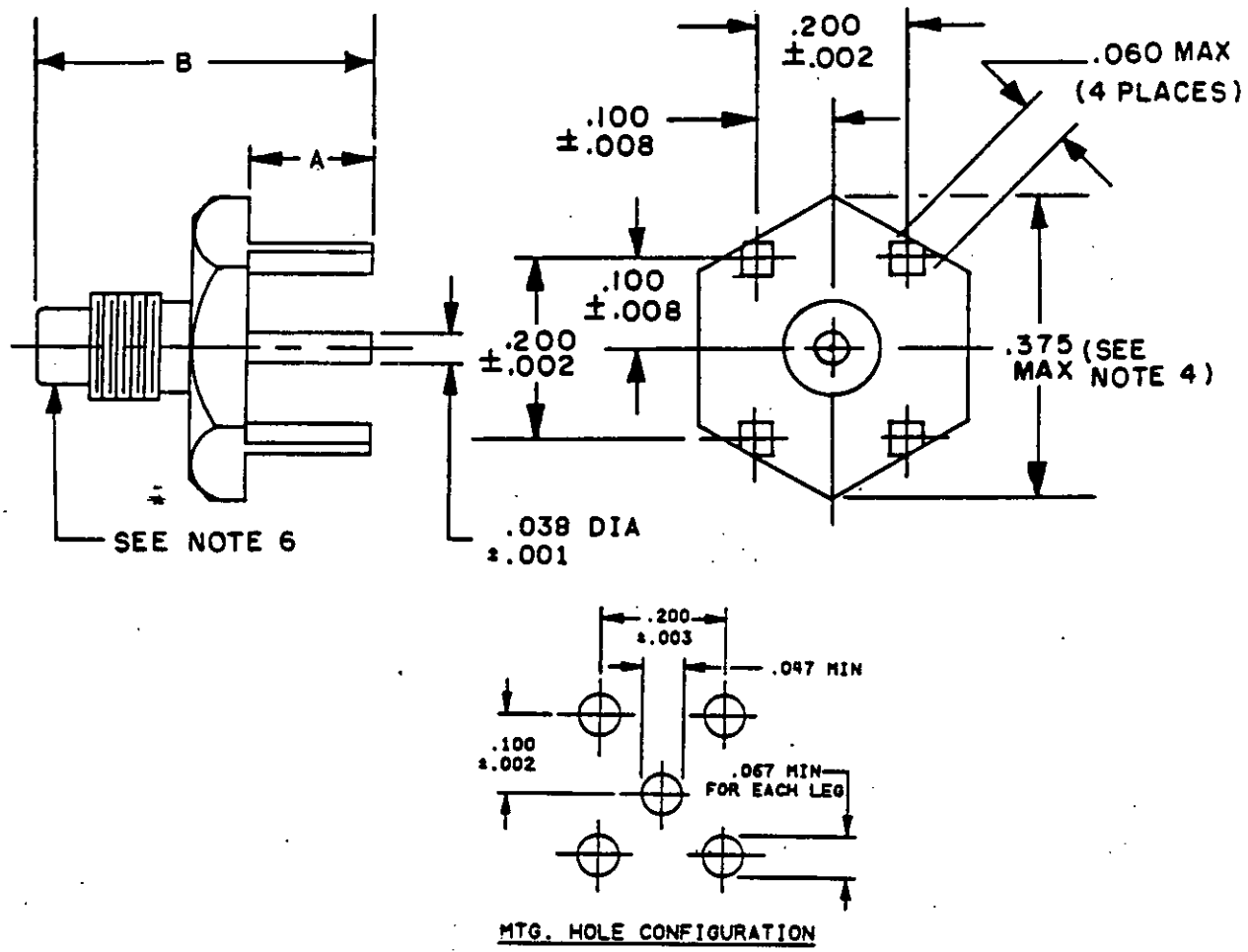


FIGURE 1. General configuration.

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Dash number	Dimension A	Dimension B
0001	.155 (3.94) ±.010 (0.25)	.455 max
0002	.125 (3.18) ±.010 (0.25)	.425 max
0003	.093 (2.36) ±.010 (0.25)	.391 max

Inches	mm	Inches	mm
.001	0.02	.200	5.08
.002	0.05	.218	5.54
.003	0.08	.367	9.32
.008	0.20	.375	9.52
.020	0.51	.385	9.78
.038	0.97	.391	9.93
.047	1.19	.425	10.80
.060	1.52	.435	11.05
.067	1.70	.455	11.56
.100	2.54	.600	15.24

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. All undimensioned pictorial representations are for reference purposes only.
4. Configuration of base optional. Dimension .375 (9.52 mm) is the maximum envelope diameter.
5. Connector bodies shall be gold plated in accordance with MIL-G-45204, type II, class I.
6. SMC interface shall be in accordance with MIL-STD-348, figure 312.1

FIGURE 1. General configuration - Continued.

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ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: 0 to 4 GHz.

Voltage rating: 335 volts rms maximum, at sea level, 85 volts rms maximum, at 70,000 feet.

Temperature rating: -65°C to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Force to engage and disengage:

Longitudinal force: Not applicable.

Torque: 16 inch-ounces maximum.

Coupling proof torque: Not applicable.

Inspection conditions: Not applicable.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance: Method 302, test condition B, MIL-STD-202. 1,000 megohms minimum.

Center contact retention:

Minimum axial force: 6.0 pounds from mating end; 4.0 pounds from opposite end.

Torque: 3 inch-ounces.

Solderability 1/: Method 208 of MIL-STD-202.

Salt spray (corrosion): Method 101, test condition B, MIL-STD-202.

Voltage standing wave ratio (VSWR): Not applicable.

Swept frequency VSWR test setup: Not applicable.

Connector durability:

Insertion and withdrawal force: 500 cycles minimum, at 12 cycles per minute maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Initial: 14 pounds maximum.

Final: 2 pounds minimum disengagement.
14 pounds maximum engage and disengage.

1/ For quality conformance inspection, the test shall be performed in group B following VSWR; test five samples with no failures permitted.

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Contact resistance: In milliohms, maximum:

	Initial	After environment
Center contact	6.0	8.0
Outer contact	1.0	1.5
Braid to body	Not applicable	Not applicable

Dielectric withstanding voltage: Method 301 of MIL-STD-202. 1,000 volts rms at sea level.

Vibration, high frequency: Method 204, test condition D, MIL-STD-202.

Shock (specified pulse): Method 213, test condition H, MIL-STD-202.

Thermal shock: Method 107, test condition B, MIL-STD-202.

Moisture resistance: Not applicable.

Corona level: Not applicable.

RF high potential withstanding voltage:

Voltage and frequency: 600 volts rms at 5 MHz.

Leakage current: Not applicable.

Cable retention force: Not applicable.

Coupling mechanism retention force: Not applicable.

RF leakage: Not applicable.

Insertion loss: Not applicable.

NOTE: This specification sheet supersedes DESC drawing 85139 when a QPL source is obtained.

Part or Identifying Number (PIN): M39012/133- (dash number from figure 1).

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 17, 99

User activities:
Army - AT
Navy - AS, MC, SH
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

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